

Interacting Multiple Model Based on Maximum Correntropy Kalman Filter

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Abstract—Traditional interacting multiple model Kalman filter (IMM-KF) was derived for linear discrete time system with Markovian coefficients and it works well under Gaussian noise. In practice, when the system is affected by the impulsive noise, the performance of the traditional IMM-KF model becomes poor. Then, maximum correntropy Kalman filter (MCKF) which has advantages in dealing with non-Gaussian noise was proposed. However, the performance of traditional MCKF becomes poor when the motion state is not single. In this brief, we propose a new algorithm called interacting multiple model based on maximum correntropy Kalman filter (IMM-MCKF) by combining interacting multiple model (IMM) with MCKF to deal with the impulsive noise. Also, the traditional MCKF works poorly when the kernel bandwidth is small, we made a further improvement by changing the kernel. Simulation examples are given to demonstrate the effectiveness of the new algorithms.

Index Terms—Maximum correntropy Kalman filter, maximum correntropy criterion, interacting multiple model, mixed-kernel, root mean square error.

I. INTRODUCTION

THE KALMAN filtering (KF) [1]–[3] is a least squares linear filter, and is widely used in target tracking [4], signal processing, system state estimation [5], navigation, control, and other fields. In recent years, as the motion models are becoming diverse, interacting multiple model Kalman filter (IMM-KF) [6]–[7] has been proposed in 1984 [8], which can work with more weighted fusion models effectively. Therefore, on the problem of maneuvering target processing, interacting multiple model (IMM) has the properties of higher precision and better stability. Now IMM-KF is widely used in maneuvering target tracking [6]–[9], positioning system [10], the navigation system [11], state estimation of the system [12], formation control [13], biomedical engineering domain [14]–[17], etc.

IMM is an algorithm for automatic recognition and switching between models based on Bayesian theory [18]. Most of the IMM-KF algorithms are based on the minimum mean

square error (MMSE) criterion which can effectively handling conventional Gaussian noise but may work poorly in the presence of non-Gaussian environments.

In order to solve this problem, maximum correntropy criterion (MCC) [19]–[23] was proposed to apply in the KF, called maximum correntropy Kalman filter (MCKF) [24], which has advantages in dealing with non-Gaussian environment and higher accuracy in tracking and positioning process. However, the performance of traditional single model MCKF becomes poor when the motion state is not single. In addition, MCKF with a small kernel bandwidth will get better steady error normally [25] and [26]. However, we find the traditional MCKF cannot work when the kernel bandwidth is small. For real maneuvering target tracking, the motion state of the target is often not single, and we pursue a smaller steady error, so it's necessary to propose a new algorithm to deal with these problems.

In this brief, we combine the conventional IMM with MCKF to propose a new algorithm, which is called interacting multiple model based on maximum correntropy Kalman filter (IMM-MCKF). Then we change its Gaussian kernel to a linear combination of Gaussian kernel and Sigmoid kernel to get better steady error. At the same time, we analyze and compare the root mean square error (RMSE) of tracking data, and it is expected that IMM-MCKF will have better performance in the environment of non-Gaussian noise in the two-state motion model than the traditional IMM-KF algorithms and traditional MCKF.

II. PRELIMINARIES

In traditional IMM-KF, suppose that there are n Kalman models. The process equation and measurement equation of model m are as follows ($m = 1, 2, \dots, n$) [8]:

$$\begin{cases} X(k+1) = F_m X(k) + w_m(k), \\ Z(k) = H_m X(k) + r_m(k), \end{cases} \quad (1)$$

where $X(k) \in \mathbb{R}^p$ represents the p -dimensional state vector at instant k , $Z(k) \in \mathbb{R}^q$ denotes the q -dimensional measurement vector at instant k , F_m is the state transition matrix of model m , and H_m is the observation transition matrix of model m . $w_m(k)$ and $r_m(k)$ are the state noise and observation noise with zero mean and covariance Q_m and R .

The Markov transition probability matrix is

$$P = \begin{bmatrix} P_{11} & \dots & P_{1n} \\ \vdots & \ddots & \vdots \\ P_{n1} & \dots & P_{nn} \end{bmatrix}, \quad (2)$$

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where P_{im} is transition probability from model i to model m .

In general, the classic IMM-KF algorithm model can be summarized into the following steps:

(a) Input interaction: Calculate the probability of mixed normalized model from model i to model m :

$$\mu_{im}(k-1|k-1) = P_{im}\mu_i(k-1)/\bar{c}_m, \quad (3)$$

where $\bar{c}_m = \sum_{i=1}^n P_{im}\mu_i(k-1)$, and $\mu_i(k-1)$ is the model probability.

(b) Calculate the initial fusion state vector of model m :

$$\hat{X}_{pre_m}(k-1|k-1) = \sum_{i=1}^n \hat{X}_i(k-1|k-1)\mu_{im}(k-1|k-1). \quad (4)$$

(c) Calculate the mixed covariance estimation of model m :

$$\begin{aligned} P_{pre_m}(k-1|k-1) &= \sum_{i=1}^n \mu_{im}(k-1|k-1) \\ &\left\{ P_i(k-1|k-1) + \left[\hat{X}_i(k-1|k-1) - \hat{X}_{pre_m}(k-1|k-1) \right] \right. \\ &\left. \left[\hat{X}_i(k-1|k-1) - \hat{X}_{pre_m}(k-1|k-1) \right]^T \right\}. \end{aligned} \quad (5)$$

(d) The KF is applied to model m :

$$\hat{X}_m(k|k) = \hat{X}_m(k|k-1) + K_m(k)v_m(k), \quad (6)$$

where

$$\begin{cases} \hat{X}_m(k|k-1) = F_m \hat{X}_{pre_m}(k-1|k-1), \\ P_m(k|k-1) = F_m P_{pre_m}(k-1|k-1) F_m^T + Q_m, \\ v_m(k) = Z(k) - H_m \hat{X}_m(k|k-1), \\ S_m(k) = H_m P_m(k|k-1) H_m^T + R, \end{cases} \quad (7)$$

$$K_m(k) = P_m(k|k-1) H_m^T S_m^{-1}(k). \quad (8)$$

(e) Observe the likelihood function of $Z(k)$:

$$\Lambda_m(k) = \frac{1}{\sqrt{2\pi|S_m(k)|}} \exp\left(-\frac{1}{2}v_m^T(k)S_m^{-1}(k)v_m(k)\right). \quad (9)$$

(f) Calculate the filtering covariance matrix:

$$\begin{aligned} P_m(k|k) &= [I - K_m(k)H_m]P_m(k|k-1) \\ &= P_m(k|k-1) - K_m(k)S_m(k)K_m^T(k). \end{aligned} \quad (10)$$

(g) Update the model probability:

$$\mu_m(k) = \frac{\Lambda_m(k)\bar{c}_m}{c}, \quad (11)$$

where

$$c = \sum_{m=1}^n \Lambda_m(k)\bar{c}_m. \quad (12)$$

(h) Output interaction: Calculate the fusion filter value:

$$\hat{X}(k|k) = \sum_{m=1}^n \hat{X}_m(k|k)\mu_m(k). \quad (13)$$

(i) Interact the covariance matrix of filtering error:

$$\begin{aligned} P(k|k) &= \sum_{m=1}^n \mu_m(k)\{P_m(k|k) \\ &+ [\hat{X}_m(k|k) - \hat{X}(k|k)][\hat{X}_m(k|k) - \hat{X}(k|k)]^T\}. \end{aligned} \quad (14)$$

III. PROPOSED IMM-MCKF

A. IMM Combined With MCKF

In the IMM-MCKF, we still assume that the observed target has n transfer equations, an interaction model with n inputs, and then calculate normalized mixed probability and covariance matrix. For the selected model m , Kalman filtering based on MCC is performed, then model probability is updated, and the interactive fusion filtering value is finally output. ($m = 1, 2, \dots, n$) Here, we refer to [8], [19] and give the new algorithm.

Suppose the model is the same as IMM-KF we mentioned above, we summarize the basic steps of the IMM-MCKF as follows.

1) Compute the Filter Value Corresponding to Each Model: (a)–(c), the same as those of IMM-KF

(d) Compute the state predicted value and the state predicted error covariance:

$$\begin{aligned} \hat{X}_m(k|k-1) &= F_m \hat{X}_{pre_m}(k-1|k-1), \\ P_m(k|k-1) &= F_m P_{pre_m}(k-1|k-1) F_m^T + Q_m. \end{aligned} \quad (15)$$

(e) Calculate the new rate and residual error:

$$\begin{aligned} v_m(k) &= Z(k) - H_m \hat{X}_m(k|k-1), \\ S_m(k) &= H_m P_m(k|k-1) H_m^T + R. \end{aligned} \quad (16)$$

(f) Preparation for calculating the filter value:

$$\begin{aligned} D_m(k) &= B_m^{-1}(k) \begin{bmatrix} \hat{X}_m(k|k-1) \\ Z(k) \end{bmatrix}, \\ W_m(k) &= B_m^{-1}(k) \begin{bmatrix} I \\ H_m(k) \end{bmatrix}, \end{aligned} \quad (17)$$

where

$$B_m(k) = \begin{bmatrix} B_{p_m}(k|k-1) & 0 \\ 0 & B_{r_m}(k) \end{bmatrix}, \quad (18)$$

and $B_{p_m}(k|k-1)$, $B_{r_m}(k)$ are obtained by Cholesky decomposition of $P_m(k|k-1)$, R :

$$\begin{aligned} P_m(k|k-1) &= B_{p_m}(k|k-1) B_{p_m}^T(k|k-1), \\ R &= B_{r_m}(k) B_{r_m}^T(k). \end{aligned} \quad (19)$$

(g) Let $t = 1$ and give initial value of the filter value for model m .

$$\hat{X}_m(k|k)_0 = \hat{X}_m(k|k-1), \quad (20)$$

where $\hat{X}_m(k|k)_t$ represents the estimated state of the iteration at t .

(h) Calculate the filter value for mode m :

$$\hat{X}_m(k|k)_t = \hat{X}_m(k|k-1) + \bar{K}_m(k)v_m(k), \quad (21)$$

where

$$\begin{aligned} \bar{K}_m(k) &= \bar{P}_m(k|k-1) H_m^T \left[H_m \bar{P}_m(k|k-1) H_m^T + \bar{R}_m(k) \right]^{-1}, \\ \bar{P}_m(k|k-1) &= B_{p_m}(k|k-1) \phi_{x_m}^{-1}(k) B_{p_m}^T(k|k-1), \\ \bar{R}_m(k) &= B_{r_m}(k) \phi_{y_m}^{-1}(k) B_{r_m}^T(k), \\ \phi_{x_m}(k) &= \text{diag}\{G_\sigma[\nabla_{1_m}(k)] \quad \dots \quad G_\sigma[\nabla_{p_m}(k)]\}, \\ \phi_{y_m}(k) &= \text{diag}\{G_\sigma[\nabla_{p+1_m}(k)] \quad \dots \quad G_\sigma[\nabla_{p+q_m}(k)]\}, \end{aligned} \quad (22)$$

with

$$\begin{aligned}\nabla_m(k) &= \mathbf{D}_m(k) - \mathbf{W}_m(k)\hat{\mathbf{X}}_m(k|k)_{t-1} \\ &= [\nabla_{1_m}(k) \quad \cdots \quad \nabla_{p+q_m}(k)]^T,\end{aligned}\quad (23)$$

and Gaussian Kernel

$$G_\sigma[\nabla] = \exp\left(-\frac{\nabla^2}{2\sigma^2}\right), \quad (24)$$

where σ is kernel bandwidth.

(i) Calculate the lower limit value in the cycle:

$$\Psi_{t_m} = \frac{|\hat{\mathbf{X}}_m(k|k)_t - \hat{\mathbf{X}}_m(k|k)_{t-1}|}{|\hat{\mathbf{X}}_m(k|k)_{t-1}|}. \quad (25)$$

If $\Psi_t \leq \varepsilon$, set $\hat{\mathbf{X}}_m(k|k) = \hat{\mathbf{X}}_m(k|k)_t$ and continue to the next step. Else, let $t = t + 1$ and back to (h).

(j) Calculate the posterior covariance matrix of the model m :

$$\begin{aligned}\mathbf{P}_m(k|k) &= [\mathbf{I} - \bar{\mathbf{K}}_m(k)\mathbf{H}_m]\mathbf{P}_m(k|k-1)[\mathbf{I} - \bar{\mathbf{K}}_m(k)\mathbf{H}_m]^T \\ &\quad + \bar{\mathbf{K}}_m(k)\mathbf{R}\bar{\mathbf{K}}_m^T(k).\end{aligned}\quad (26)$$

(k) Update the probability of the model:

$$\mu_m(k) = \frac{\Lambda_m(k)\bar{c}_m}{c}, \quad (27)$$

where

$$\begin{aligned}c &= \sum_{m=1}^n \Lambda_m(k)\bar{c}_m, \\ \Lambda_m(k) &= \frac{1}{\sqrt{2\pi}|\mathbf{S}_m(k)|} \exp\left(-\frac{1}{2}\mathbf{v}_m^T(k)\mathbf{S}_m^{-1}(k)\mathbf{v}_m(k)\right),\end{aligned}\quad (28)$$

where $\mathbf{S}_m(k)$ has been introduced in (16).

2) *Interact the Filter Value:*

$$\begin{aligned}\hat{\mathbf{X}}(k|k) &= \sum_{m=1}^n \hat{\mathbf{X}}_m(k|k)\mu_m(k), \\ \mathbf{P}(k|k) &= \sum_{m=1}^n \mu_m(k) \left\{ \mathbf{P}_m(k|k) \right. \\ &\quad \left. + [\hat{\mathbf{X}}_m(k|k) - \hat{\mathbf{X}}(k|k)][\hat{\mathbf{X}}_m(k|k) - \hat{\mathbf{X}}(k|k)]^T \right\}.\end{aligned}\quad (29)$$

B. Computational Complexity

In this sub-section, we analyze the computational complexity of IMM-MCKF in terms of the floating operations and compare it with the traditional IMM-KF.

Table I shows the computational complexities of some equations. The traditional IMM-KF mainly involves (3)-(8) and (26)-(30). We can conclude the computational complexity of IMM is

$$\begin{aligned}S_{IMM} &= [8p^3 + 9p^2q + 7pq^2 + (2n+1)p^2 - 2pq \\ &\quad + (4n-1)p + 2q^2 + q + n^2 + 3n + 4]n \\ &\quad - p^2 - 2p - 1 + O(q^3) + O(q^2).\end{aligned}\quad (31)$$

TABLE I
COMPUTATIONAL COMPLEXITIES OF SOME EQUATIONS

Equation	Multiplication and addition/subtraction	Division, matrix inversion, exponentiation, and Cholesky decomposition
(3)	$n^3 + n^2 - n$	n^2
(4)	$2pn^2 - pn$	0
(5)	$(2p^2 + 2p + 1)n^2 - (p^2 + 2p)n$	0
(8)	$2(p^2q + pq^2 - pq)n$	$O(q^3)$
(15)	$(4p^3 + p^2 - p)n$	0
(16)	$(2p^2q + 2pq^2 + pq)n$	0
(21)	$2pqn$	0
(22)	$(4p^3 + 4q^3 + 4pq^2 + 4p^2q - 2p^2 - 2q^2 - 3pq)n$	$2(p+q)n + O(q^3) + O(p^3)$
(23)	$2(p^2 + pq)n$	0
(25)	pn	n
(26)	$(4p^3 + 5p^2q + 3pq^2 - 2p^2 - 3pq)n$	0
(27)	n	n
(28)	$(2q^2 + q + 1)n - 1$	$2n + O(q^2)$
(29)	$2pn - p$	0
(30)	$3p^2n + pn - p^2 - p$	0

The IMM-MCKF mainly involves (3)-(5), (15), (16) and (21)-(30). We can also get the computational complexity of IMM-MCKF is

$$\begin{aligned}S_{IMM-MCKF} &= [(4L+8)p^3 + 4Lq^3 + (4L+7)p^2q + (4L+5)pq^2 \\ &\quad + (2n-2L+1)p^2 - Lpq + (4n+4L-1)p \\ &\quad + (2-2L)q^2 + (2L+1)q + n^2 + 3n + L + 4]n \\ &\quad - p^2 - 2p - 1 + LO(p^3) + LO(q^3) + O(q^2).\end{aligned}\quad (32)$$

The average fixed-point iteration number L is relatively small [19], so the computational complexity of IMM-MCKF is slightly larger than the traditional IMM-KF.

C. Kernel Change

In many cases, MCKF with a small kernel bandwidth performs better. However, we find the traditional MCKF cannot work when the kernel bandwidth is small. Here, we make a linear combination of Gaussian kernel and Sigmoid kernel to solve this problem.

Therefore, a new algorithm can be proposed by changing Gaussian kernel (24) into mixed-kernel (33) based on IMM-MCKF algorithm above. We called the new algorithm IMM based on improved MCKF (IMM-IMCKF). It can be expected that IMM-IMCKF has the smallest steady-state RMSEs in the following simulation compared with IMM-KF, MCKF and

IMM-MCKF.

$$L[x, y] = \alpha \exp\left(-\frac{\nabla^2}{2\sigma^2}\right) + (1 - \alpha) \tanh(\beta \Delta + \chi), \quad (33)$$

where $\nabla = \mathbf{x} - \mathbf{y}$, $\Delta = \mathbf{x} \mathbf{y}$, $0 \leq \alpha \leq 1$ is the mixing coefficient, and β, χ are Sigmoid kernel parameters.

IV. SIMULATIONS

In this part, we do a tracking simulation combining constant velocity (CV) with constant turning (CT) model, and compare the results under CV single model MCKF (MCKF-CV), traditional IMM-KF, IMM-MCKF and IMM-IMCKF. The target, whose initial location is (1000, 1000)/(m) and initial speed is (10, -10)/(m/s), first moves in a straight line, turns at a uniform angular speed, and then moves in a straight line. The simulation results are averaged over 500 Monte Carlo runs. Here, we refer to the simulation conditions in [9].

First of all, we consider the motion model of the two states, so here $m=2$, and the linear system is as follows:

$$\begin{cases} \mathbf{X}(k+1) = \mathbf{F}_m \mathbf{X}(k) + \mathbf{w}_m(k), \\ \mathbf{Z}(k) = \mathbf{H}_m \mathbf{X}(k) + \mathbf{r}_m(k), \end{cases} \quad (34)$$

where $\omega = \frac{\pi}{270}$,

$$\mathbf{F}_1 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{F}_2 = \begin{bmatrix} 1 & \frac{\sin(\omega)}{\omega} & 0 & \frac{\cos(\omega)-1}{\omega} \\ 0 & \cos(\omega) & 0 & -\sin(\omega) \\ 0 & \frac{1-\cos(\omega)}{\omega} & 1 & \frac{\sin(\omega)}{\omega} \\ 0 & \sin(\omega) & 0 & \cos(\omega) \end{bmatrix},$$

$$\mathbf{H}_1 = \mathbf{H}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{X}(0) = \begin{bmatrix} 1000 \\ 10 \\ 1000 \\ -10 \end{bmatrix}, \quad (35)$$

and the Markov transition probability matrix is

$$\mathbf{P} = \begin{bmatrix} 0.99 & 0.01 \\ 0.01 & 0.99 \end{bmatrix}. \quad (36)$$

Secondly, we consider that the noise in the measurement equation is impulse noise whose distribution is mixed-Gaussian [19]:

$$\begin{aligned} \mathbf{r}_1(k) &\sim 0.9N(0, 0.01^2) + 0.1N(0, 200^2), \\ \mathbf{r}_2(k) &\sim 0.9N(0, 0.01^2) + 0.1N(0, 200^2), \end{aligned} \quad (37)$$

and the noise in the process equation is Gaussian:

$$\begin{aligned} \mathbf{w}_1(k) &\sim \mathbf{G}_1 N(0, \mathbf{Q}_1), \\ \mathbf{w}_2(k) &\sim \mathbf{G}_2 N(0, \mathbf{Q}_2), \end{aligned} \quad (38)$$

where

$$\mathbf{G}_1 = \begin{bmatrix} \frac{1}{2} & 0 \\ 1 & 0 \\ 0 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}, \quad \mathbf{G}_2 = \begin{bmatrix} \frac{1}{2} & 0 \\ 1 & 0 \\ 0 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}, \quad (39)$$

$$\mathbf{Q}_1 = \begin{bmatrix} 0.1^2 & 0 \\ 0 & 0.1^2 \end{bmatrix}, \quad \mathbf{Q}_2 = \begin{bmatrix} 0.0144^2 & 0 \\ 0 & 0.0144^2 \end{bmatrix}. \quad (40)$$

In Figure 1, we set kernel bandwidth $\sigma = 3$ (it cannot be smaller or the algorithm will not work), and the loop restriction $\varepsilon = 0.01$ in MCKF-CV and IMM-MCKF, $\alpha = 0.9$, $\sigma_I = 1.1$,

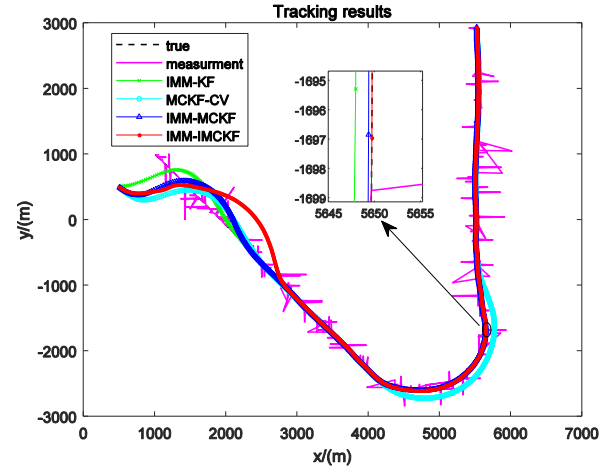


Fig. 1. Tracking results under four different filters.

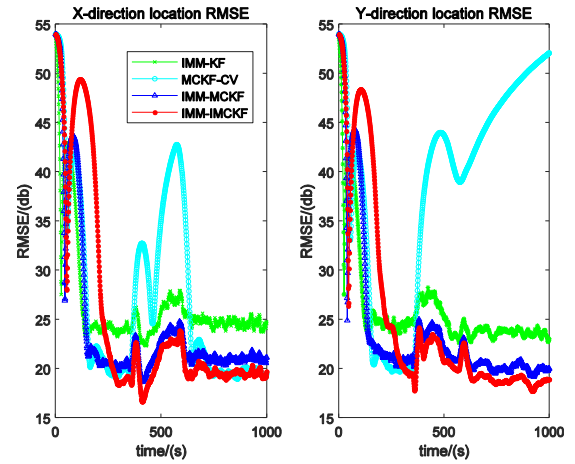


Fig. 2. RMSEs of tracking under four different filters.

TABLE II
RMSES OF X-DIRECTION, Y-DIRECTION IN NON-GAUSSIAN NOISES
UNDER FOUR DIFFERENT ALGORITHMS

Filters	RMSE of x-direction/(m)	RMSE of y-direction/(m)
MCKF-CV	32.7028	359.6263
IMM-KF	17.9187	16.3698
IMM-MCKF	11.9316	11.1056
IMM-IMCKF	10.1435	9.6491

$\beta = 0.2$, $\chi = 5$ in IMM-IMCKF and it shows the tracking results in non-Gaussian noises under these four algorithms. We can clearly see that MCKF-CV can't work well when the motion state of target changes, IMM-KF, IMM-MCKF and IMM-IMCKF both doing better than MCKF-CV.

Figure 2 shows RMSEs of x-direction, y-direction under IMM-KF, IMM-MCKF, IMM-IMCKF and MCKF-CV in non-Gaussian noise as (37). IMM-KF converges faster than IMM-MCKF and IMM-IMCKF. But IMM-MCKF and IMM-IMCKF have better steady-state error.

Then we present in Table II specific numerical comparisons of the steady RMSEs in x-direction and y-direction under different algorithms. The numbers show that IMM-IMCKF is the best one compared with other three algorithms in this situation.

TABLE III
RMSES IN NON-GAUSSIAN NOISES WITH DIFFERENT KERNEL
BANDWIDTH UNDER IMM-MCKF AND IMM-IMCKF

Bandwidth	(RMSE of x-direction/(m), RMSE of y-direction/(m))	
	IMM-MCKF	IMM-IMCKF
$\sigma=10$	(16.29, 14.66)	(16.41, 14.78)
$\sigma=8$	(15.97, 14.90)	(16.14, 15.07)
$\sigma=6$	(14.83, 14.02)	(15.10, 14.25)
$\sigma=4$	(13.00, 12.32)	(13.36, 12.68)
$\sigma=2$	Cannot work	(10.89, 10.37)
$\sigma=1$	Cannot work	(10.28, 9.68)

TABLE IV
RMSES IN GAUSSIAN NOISES UNDER FOUR FILTERS

Filters	RMSE of x-direction/(m)	RMSE of y-direction/(m)
MCKF-CV	10.7732	11.5062
IMM-KF	7.1073	6.7424
IMM-MCKF	7.1047	6.7461
IMM-IMCKF	7.1049	6.7457

And the RMSEs of IMM-IMCKF are about 40 percent lower compared with IMM-KF.

In Table III, we changed the kernel bandwidth and observed the RMSEs of IMM-MCKF, IMM-IMCKF in x-direction and y-direction. When σ gets smaller, the RMSEs become better. However, IMM-MCKF cannot work when σ is too small. At the same time, IMM-IMCKF still work well and get better results.

Then we consider Gaussian noise in the simulation,

$$\begin{aligned} r_1(k) &\sim N(0, 2\sigma^2), \\ r_2(k) &\sim N(0, 2\sigma^2), \end{aligned} \quad (41)$$

Table IV shows the performance of our proposed system under gaussian noise, where we set kernel bandwidth $\sigma = 10$, the loop restriction $\varepsilon = 0.01$. IMM-KF, IMM-MCKF, IMM-IMCKF have almost identical steady RMSE. And they perform better than MCKF-CV.

V. CONCLUSION

A new IMM algorithm based on MCKF called IMM-MCKF has been proposed. It has better tracking ability against impulsive noise by changing the filter gain of the original IMM-KF algorithm. Simulation results demonstrate that IMM-MCKF can outperform the traditional IMM-KF and MCKF in the two-state motion model in the non-Gaussian noises, even though the computational complexity of IMM-MCKF is slightly larger than the traditional IMM-KF. Then further improvement has been made by changing kernel. And it is useful for IMM-MCKF, IMM-IMCKF to select an appropriate kernel bandwidth.

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